

DUNGEON QUEST RPG

Final Project Documentation

1. Project Introduction

Dungeon Quest RPG is a simple 2D Role-Playing Game (RPG) developed using Java. The main idea of the game is to allow the player to control a character, explore a game environment, fight enemies, complete a quest, upgrade the weapon, and finally defeat a powerful boss.

The project was developed step by step. At the beginning, the basic player movement and game environment were created. Later, combat, enemies, health, quests, XP, weapon upgrades, boss battles, sound effects, and visual improvements were added.

In the final stage, the game was tested and improved to make it more stable and enjoyable.

The main goal of the project was to create a small but complete RPG game using Java and Java Swing while applying programming concepts such as classes, objects, methods, conditions, loops, event handling, and game logic.

2. Project Objectives

The main objectives of the Dungeon Quest RPG project are:

1. To develop a simple and playable RPG game using Java.
2. To create a controllable player character.
3. To implement an enemy and combat system.
4. To create a boss battle system.
5. To develop a quest and progress system.
6. To add a health and XP system.
7. To provide a weapon upgrade feature.
8. To improve the visual appearance of the game.
9. To add sound effects and background music.
10. To create a proper game completion system.
11. To practice object-oriented programming concepts.
12. To test and prepare a final playable version of the game.

3. Game Features

The final version of Dungeon Quest RPG contains several gameplay features.

3.1 Player Movement

The player can move around the game environment using the keyboard.

The controls are:

Key	Function
W	Move Up
S	Move Down
A	Move Left
D	Move Right
SPACE	Attack
U	Upgrade Weapon

The movement system allows the player to explore the game area and move toward enemies.

3.2 Combat System

The game contains a simple combat system.

The player can attack nearby enemies using the **SPACE** key. When an enemy is attacked, its health decreases.

When an enemy's health reaches zero, the enemy is defeated.

The combat system makes the game interactive and gives the player a clear objective during gameplay.

3.3 Enemy System

Multiple enemies are placed throughout the game environment.

Each enemy has its own position and health. The player needs to find and defeat the enemies to make progress in the game.

Defeating enemies also provides experience points to the player.

3.4 Boss Battle

A special boss character is included in the game.

The boss is stronger than the normal enemies and provides the final major challenge for the player.

The player needs to defeat the boss to complete the main objective of the game.

The boss also has a larger health value and a different visual appearance from normal enemies.

3.5 Quest System

The game contains a quest system that keeps track of the player's progress.

The player needs to defeat the required enemies and complete the necessary objectives before the game can be completed.

The quest system helps give the gameplay a clear direction instead of allowing the player to fight enemies without a purpose.

3.6 XP System

The game includes an experience point system.

When the player defeats enemies, the player receives XP.

The XP system gives the player a sense of progress during the game.

3.7 Weapon Upgrade

The player can upgrade the weapon during gameplay.

The U key is used for the weapon upgrade.

The upgrade system allows the player to improve their attack ability and provides an additional gameplay feature.

3.8 Game Environment

The final game environment was improved from the basic version.

The environment contains:

- Green grass
- Trees
- Bushes
- Rocks
- Flowers
- Water
- Paths
- Obstacles

These elements make the game area look more like an RPG environment and make the gameplay area more interesting.

3.9 Sound System

Sound was added to improve the gaming experience.

The game uses different sound files for different situations, including:

- Background music
- Fight sound
- Enemy defeat/event sound

The sounds provide additional feedback during gameplay.

3.10 Game Completion System

A proper completion system was added to the final version.

The game checks whether the required enemies and boss have been defeated.

After the objectives are completed, the game enters the **Game Complete** state and displays the completion screen.

This provides a clear ending for the player.

4. Technologies Used

The main technologies and tools used to develop the project are:

Technology/Tool	Purpose
Java	Main programming language
Java Swing	Game window and graphical interface
Java AWT	Graphics, keyboard events and game components
VS Code	Code development and project management
WAV Audio Files	Background and gameplay sounds

Java was selected because it supports object-oriented programming and provides useful libraries for creating a 2D graphical game.

5. Gameplay System

The basic gameplay process is:

Start Game → Explore → Find Enemies → Attack → Defeat Enemies → Gain XP → Upgrade Weapon → Complete Quest → Fight Boss → Defeat Boss → Game Complete

Step 1: Start the Game

The player starts the Dungeon Quest RPG and enters the game environment.

Step 2: Explore the Environment

The player can move around the game area using W, A, S, and D.

Step 3: Fight Enemies

The player finds enemies and uses the SPACE key to attack them.

Step 4: Gain Experience

After defeating enemies, the player receives XP.

Step 5: Upgrade Weapon

The player can use the U key to upgrade the weapon.

Step 6: Complete the Quest

The player continues defeating the required enemies to progress through the quest.

Step 7: Boss Battle

After completing the required objectives, the player faces the boss.

Step 8: Complete the Game

After defeating the boss, the game shows the Game Complete screen.

6. Development Process

The Dungeon Quest RPG was developed gradually over several weeks.

Initial Development

At the beginning, the basic game window, player, movement, and game environment were created.

Gameplay Development

After that, the main gameplay systems were added:

- Player health
- Enemy system
- Attack system
- Weapon system
- Quest system
- XP system

Boss System

A separate boss battle system was then added. The boss was designed to be stronger than the normal enemies and provide a final challenge.

Visual Improvement

The basic game environment was improved with grass, trees, bushes, rocks, flowers, water, paths, and other details.

The player and enemies were also given more recognizable designs.

Audio Improvement

Sound effects and background music were added to make the game more interactive.

Final Improvement

In the final weeks, the game was tested, bugs were fixed, the code was organized, and the final game completion system was checked.

7. Project Source Code Structure

The project uses multiple Java classes to keep different parts of the game organized.

Main.java

This class starts the application and launches the game.

GameFrame.java

This class creates the main game window and adds the game panel.

GamePanel.java

This is the main gameplay class. It manages the game screen, player actions, enemies, game updates, keyboard controls, graphics, and game completion.

Player.java

This class manages the player character and player-related information such as movement, health, attack, XP, and weapon upgrades.

Enemy.java

This class manages the normal enemy characters.

Boss.java

This class manages the special boss character.

Dungeon.java

This class manages the dungeon environment and enemies.

Quest.java

This class manages quest progress and objectives.

AudioManager.java

This class manages background music and sound effects.

8. Testing

Testing was performed before preparing the final version of the game.

Different parts of the game were checked separately and together.

Player Testing

The following were tested:

- Player movement
- Player attack
- Player health
- Weapon upgrade
- Keyboard controls

Enemy Testing

The following were tested:

- Enemy placement
- Enemy health
- Enemy attack
- Enemy defeat
- Enemy counting

Boss Testing

The following were tested:

- Boss appearance
- Boss health
- Boss battle
- Boss defeat
- Boss completion condition

Quest Testing

The quest system was tested to make sure that enemy defeats and boss defeat were properly connected with the completion condition.

Sound Testing

Background music and gameplay sound effects were tested to make sure they played during the correct actions.

Final Completion Testing

The complete game was played from the beginning to the end to check whether the final Game Complete screen appeared correctly after completing the required objectives.

9. Final Improvements

During the final improvement stage, several parts of the game were enhanced.

Graphics

The game environment was improved with more objects and details such as trees, rocks, flowers, bushes, grass, water, and paths.

Interface

The HUD was improved to show important information clearly.

Characters

The player, enemies, and boss received improved visual designs.

Gameplay

The combat, quest, XP, weapon upgrade, and boss systems were checked and improved.

Audio

Background music and action sounds were added to improve the gameplay experience.

Completion

A proper Game Complete screen was added so that the player receives a clear ending.

10. Deployment

After the final testing and improvements, the project was prepared as the final playable version.

The final project contains the Java source files and required sound resources.

The game can be launched through the main Java program and played using the keyboard controls.

The final deployment version includes all major gameplay systems developed throughout the project.

11. Final Game Showcase

The final project will be demonstrated through a **live gameplay showcase**.

During the showcase, the complete game can be played from the beginning to demonstrate its major features.

The showcase will include:

1. Starting the game.
2. Moving the player around the environment.
3. Exploring the game area.
4. Fighting enemies.
5. Defeating enemies.
6. Showing the XP system.
7. Upgrading the weapon.
8. Completing the quest.
9. Fighting the boss.
10. Defeating the boss.
11. Showing the Game Complete screen.

This allows the complete project to be demonstrated through actual gameplay.

12. Final Result

The final result of the project is a complete playable **2D RPG game** developed using Java.

The game successfully combines:

- Player movement
- Combat
- Enemies
- Boss battle
- Quest system
- XP system
- Weapon upgrade
- Health system
- Graphics
- Sound
- Game environment
- Game completion system

The project started with basic gameplay and gradually developed into a more complete and visually improved RPG game.

13. Conclusion

The **Dungeon Quest RPG** project provided practical experience in game development and Java programming.

Throughout the development process, different programming concepts were used to create the player, enemies, combat, quests, boss battle, sound system, and game environment.

The final stages focused on improving the graphics, organizing the code, preparing documentation, testing the complete game, fixing problems, and preparing the final playable version.

The completed game provides a simple RPG experience where the player explores the environment, defeats enemies, gains experience, upgrades the weapon, completes the quest, and finally defeats the boss.

The final **Dungeon Quest RPG** is ready to be showcased as a playable Java-based RPG project.